THREE-DIMENSIONAL AIR POLLUTANT MODELING IN THE LOWER ATMOSPHERE

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Abstract. A steady-state, three-dimensional turbulent diffusion equation describing the concentration distribution of an air pollutant from an elevated point source in the lower atmosphere is solved analytically. The same formulation can be used to obtain solutions from line, area or other kinds of sources. The solution is developed for the cases in which the velocity, vertical and lateral diffusivities are given by the power law. The model preserves the beauty of analytical solution without sacrificing much on the accuracy of approximating the velocity and eddy diffusivities. Methods of evaluating the parameters, which are required for the model applications, are discussed. Results indicate that the ratio of the plume width to the plume length increases with decreasing stability and with increasing source height. These consequences are in response to the variations of the size of eddies in the vertical direction.

1. Introduction

Two-dimensional diffusion equations, when the velocity and the vertical diffusivity are given as power functions of the distance from the surface, have been extensively treated elsewhere (Sutton, 1934; Sutton, 1943; Frost, 1946; Calder, 1949; Yih, 1952a; Yeh and Brutsaert, 1970, 1971). Three-dimensional diffusion, where lateral diffusivity must be considered, has also been treated by several authors (Davies, 1947, 1950; Sutton, 1953; Yih, 1952b; Smith, 1957). Davies (1947, 1950) and Sutton (1953) assumed that the velocity was independent of the height in order to obtain a closed form solution. While Yih (1952b) allowed the velocity to be given by a power law, the solution was valid only for a point source situated at the origin. Thus, it cannot be used to integrate over a line or area source, which is more likely to be the case in nature.

The present paper, therefore, is concerned with the solution of the diffusion from a point source situated anywhere in the region of interest. The solution for any other kind of sources can then be obtained by integrating over the point-source solution. The basic equation that will be solved for this purpose is a steady, three-dimensional advective turbulent diffusion equation,

$$u\frac{\partial c}{\partial x} = \frac{\partial}{\partial y}\left(K_y\frac{\partial c}{\partial y}\right) + \frac{\partial}{\partial z}\left(K_z\frac{\partial c}{\partial z}\right) + S(x, y, z), \tag{1}$$

where c is the concentration; u is the wind velocity; x, y, and z are the longitudinal, lateral and vertical coordinates, respectively; K_y and K_z are the lateral and vertical diffusivities; and S is a source function.

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Theoretical models are available to determine the velocity, u, and the eddy diffusivities, K_y and K_z, as functions of the vertical coordinate, z (Ellison, 1957; Yamamoto, 1959; Panofsky, 1961; Kondo, 1962; Swinbank, 1964). However, the resulting functions are such that they make the analytical solution of Equation (1) under appropriate boundary conditions extremely difficult, if not impossible. Fortunately, it has been shown that the power law approximation of the velocity and diffusivities are suitable for the diffusion modeling in the lower atmosphere (Brutsaert and Yeh, 1970a, 1970b). Therefore, the following approximations will be made:

$$u = az^{\alpha}, \tag{2}$$

$$K_z = bz^{\beta}, \tag{3}$$

and

$$K_y = B(x)\, z^{\gamma}, \tag{4}$$

where a, b, α, β and γ are constants depending on the surface roughness and the atmospheric stability conditions (Brutsaert and Yeh, 1970a, 1970b), and B may be any integrable function of x.

Boundary conditions for Equations (1) through (4) are specified from physical consideration as follows:

$$c = c_a \quad \text{at} \quad x = 0, \tag{5}$$

$$c = c_a \quad \text{at} \quad y = \pm\infty, \tag{6}$$

$$c = c_a \quad \text{at} \quad z = \infty, \tag{7}$$

and

$$K_z \frac{\partial c}{\partial z} = 0 \quad \text{at} \quad z = 0, \tag{8}$$

where c_a is the concentration of the ambient fluid. Condition (8) stipulates that the wall is impervious to the quantity under diffusion. Conditions (7) and (6) state that the concentration of the substance tends to its ambient value at infinity, and condition (5) implies that the incoming fluid has a constant concentration.

2. Method of Solution

The advective diffusion equation is formally equivalent to the heat conduction equation if the coefficients, u, K_y and K_z, are constants. Many of the fundamental mathematical techniques were developed in the field of heat transfer, and the most extensive account of methods of solving this type of equation was given by Carslaw and Jaeger (1959). In the following, methods of Fourier transformation and Green's theorems are adopted to solve Equation (1) subject to boundary conditions (5) through (8).

Let $\tilde{\chi}(x, \omega, z)$ be the Fourier transformation of χ with respect to y, i.e.,

$$\tilde{\chi} = \int_{-\infty}^{\infty} e^{-i\omega y} \chi(x, y, z)\, \mathrm{d}y, \tag{9}$$

where $\chi = c - c_a$. Taking the Fourier transformation of Equations (1) and (5) through (8), the following system of equations yields for the case of $\alpha = \beta$ (Yih, 1952b):

$$\frac{\partial}{\partial z}\left(bz^\beta \frac{\partial \tilde{\phi}}{\partial z}\right) - az^\alpha \frac{\partial \tilde{\phi}}{\partial x} = -\tilde{S}e^{\omega^2 X/a}, \tag{10}$$

$$\tilde{\phi} = 0 \quad \text{at} \quad x = 0, \tag{11}$$

$$\tilde{\phi} = 0 \quad \text{at} \quad z = \infty, \tag{12}$$

and

$$bz^\beta \frac{\partial \tilde{\phi}}{\partial z} = 0 \quad \text{at} \quad z = 0, \tag{13}$$

where $\tilde{S}$ is the Fourier transformation of S and $\tilde{\phi}$ is defined as

$$\tilde{\phi} = \tilde{\chi} e^{\omega^2 X/a}. \tag{14}$$

In Equation (14), X is defined as

$$X = \int_0^x B(t)\,\mathrm{d}t. \tag{15a}$$

The solution of Equation (10) can be expressed in terms of its corresponding Green's function, $G(x, z; x_0, z_0)$, (Morse and Feshbach, 1953; Yeh, 1975)

$$\tilde{\phi}(x, \omega, z) = \int_0^x \int_0^\infty G\tilde{S}(x_0, \omega, z_0)\, e^{\omega^2 X_0/a}\,\mathrm{d}x_0\,\mathrm{d}z_0, \tag{16}$$

where X_0 denotes

$$X_0 = \int_0^{x_0} B(t)\,\mathrm{d}t. \tag{15b}$$

Substituting Equation (14) into Equation (16) and taking the inverse Fourier transformation on the resulting $\tilde{\chi}$, one obtains the following solution

$$\chi(x, y, z) = \frac{1}{2\pi} \int_0^\infty \int_0^x \int_{-\infty}^\infty e^{i\omega y - \omega^2 (X - X_0)/a} G\tilde{S}(x_0, \omega, z_0)\,\mathrm{d}\omega\,\mathrm{d}x_0\,\mathrm{d}z_0. \tag{17}$$

The Green's function for the type of Equations (10) through (13) has been constructed explicitly (Yeh, 1975),

$$G = \begin{cases} \dfrac{(zz_0)^{(1-\beta)/2}}{b(2+\alpha-\beta)(x-x_0)} \exp\left[-\dfrac{a\left(z^{2+\alpha-\beta} + z_0^{2+\alpha-\beta}\right)}{b(2+\alpha-\beta)^2 (x-x_0)}\right] \times \\ I_{-\nu}\left[\dfrac{2a(zz_0)^{(2+\alpha-\beta)/2}}{b(2+\alpha-\beta)^2 (x-x_0)}\right] & \text{for} \quad x > x_0, \\ 0 & \text{for} \quad x < x_0, \end{cases} \tag{18}$$

where $\nu=(1-\beta)/(2+\alpha-\beta)$. Equation (18) is valid for $\nu<1$, which is normally encountered in the atmospheric diffusion study.

For a point source located at (x_s, y_s, z_s), S can be defined as

$$S=M\,\delta(x-x_s)\,\delta(y-y_s)\,\delta(z-z_s), \tag{19a}$$

so that

$$S(x_0, z_0)=\mathrm{M}e^{-i\omega\eta_s}\delta(x_0-x_s)\,\delta(z_0-z_s), \tag{19b}$$

where M is the rate of mass released at the source point. Substituting Equations (18) and (19) into Equation (17), one obtains the following solution after performing the integration,

$$\chi=M\frac{\sqrt{a}}{\sqrt{4\pi(X-X_s)}}\exp\left[-\frac{a(y-y_s)^2}{4(X-X_s)}\right]\frac{(zz_s)^{(1-\beta)/2}}{b(2+\alpha-\beta)(x-x_s)}\times$$
$$\exp\left[-\frac{a(z^{2+\alpha-\beta}+z_s^{2+\alpha-\beta})}{b(2+\alpha-\beta)^2(x-x_s)}\right]I_{-\nu}\left[\frac{2a(zz_s)^{(2+\alpha-\beta)/2}}{b(2+\alpha-\beta)^2(x-x_s)}\right], \tag{20}$$

where X_s is defined as

$$X_s=\int_0^{x_s}B(t)\,\mathrm{d}t. \tag{15c}$$

Equation (20) is the main result of this paper for the case of a point source. The solution for other kinds of sources can be obtained either by integrating Equation (17) in conjunction with Equation (18) and the source function specified in a similar manner to Equation (19), or by integrating Equation (20) with respect to x_s, y_s and z_s over the domain of the source.

To see the generality of the present solution, it is shown that some special cases can be obtained from Equation (20). First, for constant velocity and diffusivities, $\alpha=0$, $\beta=0$ and $\nu=\frac{1}{2}$, Equation (20) is reduced to

$$\chi=M\frac{1}{\sqrt{4\pi(X-X_s)}}\exp\left[-\frac{a(y-y_s)^2}{4(X-X_s)}\right]\frac{1}{\sqrt{4\pi(x-x_s)}}\times$$
$$\left\{\exp\left[-\frac{a(z-z_s)^2}{4b(x-x_s)}\right]+\exp\left[-\frac{a(z+z_s)^2}{4b(x-x_s)}\right]\right\}, \tag{21}$$

which, with $X=x$ and $X_s=x_s$, is the well-known Gaussian diffusion model. Second, for the point source at origin, $(x_s, y_s, z_s)=(0, 0, 0,)$ Equation (20) becomes

$$\chi=M\frac{\sqrt{a}}{\sqrt{4\pi X}}\,\mathrm{Exp}\left[-\frac{ay^2}{4X}\right]\frac{1}{\Gamma(1-\nu)}\left(\frac{b}{a}\right)^{\nu}\frac{1}{b(2+\alpha-\beta)^{1-2\nu}}\frac{1}{x^{1-\nu}}\times$$
$$\times\exp\left[-\frac{az^{2+\alpha-\beta}}{b(2+\alpha-\beta)^2x}\right], \tag{22}$$

because of

$$\lim_{x\to 0} I_\nu(x) = (x/2)^\nu/\Gamma(1+\nu) \quad \text{for} \quad \nu \neq -1, -2, -3, \ldots.$$

Equation (22) was the solution obtained by Yih (1952b).

3. Power Law Approximations and the Selection of Parameters

Equation (20) describes the increase of concentration over the ambient value resulting from a continuous point source release in the lower atmosphere with variable velocity and diffusivity profiles. Its validity depends on the power law approximations for the velocity and diffusivities in the lower atmosphere. Its application to practical problems requires the specifications of the parameters, a, b, α, and β, and the function $B(x)$.

It has been shown that the power law approximations for the wind velocity and eddy diffusivities are indeed good ones in the lower atmosphere by matching the power functions with the theoretically derived wind functions (Brutsaert and Yeh, 1970a). These approximations have been applied successfully to the prediction of evaporation, which is diffusion from ground sources (Brutsaert and Yeh, 1970b; Yeh and Brutsaert, 1970, 1971). They are definitely better approximations than, and hence have the advantage over, the widely adopted assumption of constant velocity and diffusivities (Gifford, 1961, 1968; Hanna, 1974). Of course, more accurate representation of the velocity and diffusivities in the lower atmosphere than the power law can be obtained elsewhere (Swinbank, 1968; Laikhtman and Panomareva, 1969; Dyer and Hicks, 1970; Webb, 1970; Businger *et al.*, 1971; Pruitt *et al.*, 1973). However, adopting those functions, the beauty of analytical solution of Equation (1) would perhaps have to be abandoned and the time-consuming and tedius numerical techniques would have to be resorted to (Yamamoto and Shimanuki, 1960; Ragland and Peirce, 1975). On the other hand, the conventional air pollution models using the Gaussian distribution (Gifford, 1961, 1968; Hanna, 1974) preserve the beauty of analytical solution but at the expense of unrealistically representing the wind velocity and eddy diffusivities as constants. The present model offers, in the author's opinion, the best compromise. It preserves the beauty of analytical solution without sacrificing much on the accuracy of approximating the velocity and eddy diffusivities.

For model applications, the proper choices of a, b, α, β and B are essential. In atmospheric modeling, a, b, α, and β depend on the surface roughness and atmospheric stability conditions. In the absence of wind and diffusivity profiles, they can be determined from routine meteorological observations. Methods of evaluating a, b, α, and β have already been published when Reynolds' analogy is applicable (Brutsaert and Yeh, 1970a, 1970b),

$$\beta = 1 - \alpha, \tag{23a}$$

$$a = Cu_*/z_0^\alpha, \tag{23b}$$

$$b = u_* z_0^\alpha/(\alpha C), \tag{23c}$$

where u_* is the friction velocity, z_0 is the surface roughness and C is a parameter. The values of α and C have been tabulated in Brutsaert and Yeh's paper (1970a) for values of z_0/L ranging from $-8\cdot10^{-3}$ to $3\cdot10^{-5}$, where L is the well-known Monin-Obukhov stability length. Thus, knowing the stability condition of the atmosphere, L, the surface roughness, z_0, and the friction velocity, u_*, one can determine the parameters, a, b, α and β, from the tables given by Brutsaert and Yeh (1970a) and Equation (23). If Reynolds' analogy is not applicable, these parameters must be obtained from field data.

The determination of the function, $B(x)$, requires further elaboration. Smith (1957) found that the rate of decrease in the surface concentration along the x-axis is $\chi_0 \sim x^{-5/3}$ under neutral conditions, i.e., $\alpha=\frac{1}{7}$ and $\beta=\frac{6}{7}$. Yamamoto and Shimanuki (1960) solved the three-dimensional turbulent diffusion equation numerically and found that, under neutral conditions, $\chi_0 \sim x^{-1.78}$, which agrees well with existing empirical data. If one assumes $B(x) \sim x^{2/3}$, one will obtain $\tilde{\chi}_0 \sim x^{-1.72}$ from Equation (20) for the case with $\alpha=\frac{1}{7}$ and $\beta=\frac{6}{7}$. The index -1.72 falls between $-\frac{5}{3}$ and -1.78. Hence, it is proposed in this paper that the function, $B(x)$, be given by

$$B(x)=kx^p, \tag{24}$$

where k and p are two constants. These constants will be presumed to depend on the atmospheric stability and surface roughness. Under neutral conditions, p is equal to $\frac{2}{3}$ from the above discussion. Under adiabatic conditions, the values of k and p warrant further research.

4. Source Height and Stability Effects

In order to assess the effects of the source height and atmospheric stability on the distribution of concentrations, three different heights and three stability classes are considered to illustrate these effects. The results are shown in Figures 1, 2 and 3. The comparisons of concentration isopleths from a point source at $x_s=0, y_s=0$ and $z_s=3$ m with the source strength of 0.235 gram s^{-1} are shown in Figure 1 for stable, neutral and unstable conditions, respectively. Table 1 lists the parameters used for three different atmospheric stability conditions. Figures 2 and 3 depict the concentration distribution when the source is placed at $x_s=0$, $y_s=0$, $z_s=15$ m and $x_s=0$, $y_s=0$, $z_s=30$ m, respectively. With the exception of source location, all parameters used for Figures 2 and 3 are the same as for Figure 1. From these figures, it is seen that the differences in the pattern of concentration distribution between the unstable condition and the neutral and stable conditions are very pronounced. Under neutral and stable conditions, the distributions resemble the Gaussian patterns in shape. Under unstable conditions, however, the non-Gaussian shape in the z direction prevails. This non-Gaussian behavior cannot be obtained by the conventional air pollution models (Gifford, 1961, 1968; Hanna, 1974). It is also seen that the distribution patterns are quite different as the source moves from the ground to a higher elevation.

When the source is near the ground, the plume width in the vertical is much smaller than the plume length in the horizontal. The small sizes of the eddies in the lower

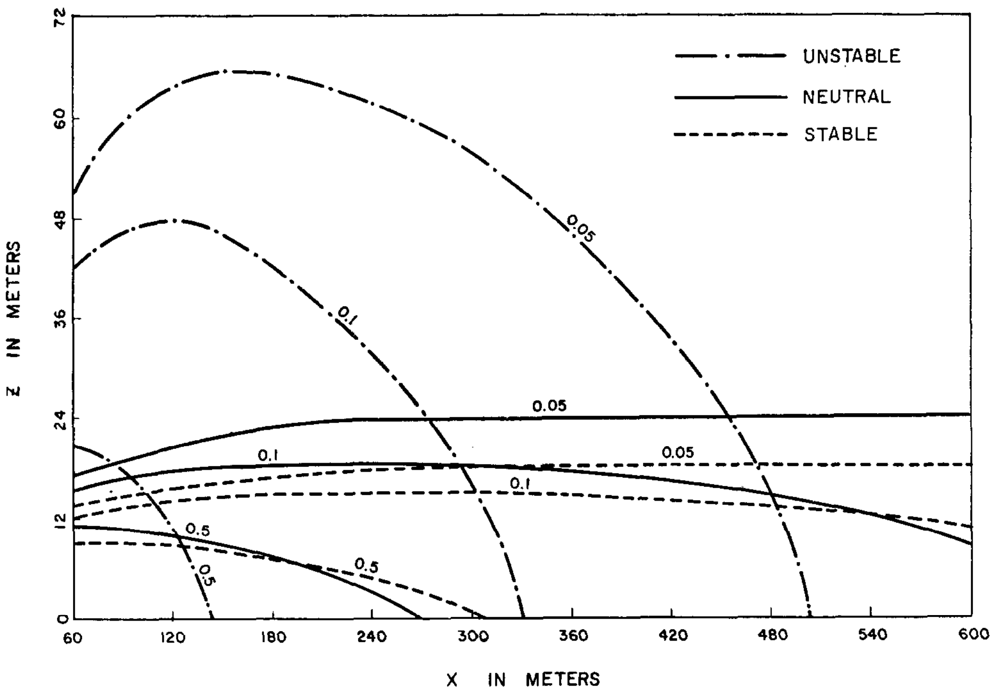

Fig. 1. Comparison of iso-concentration lines on the plane $y=0$ m for the case of a point source of strength 0.235 gram s^{-1} at $x_s=0$, $y_s=0$ and $z_s=3$ m under stable, neutral and unstable conditions, respectively. The unit of concentration is in mg m^{-3}.

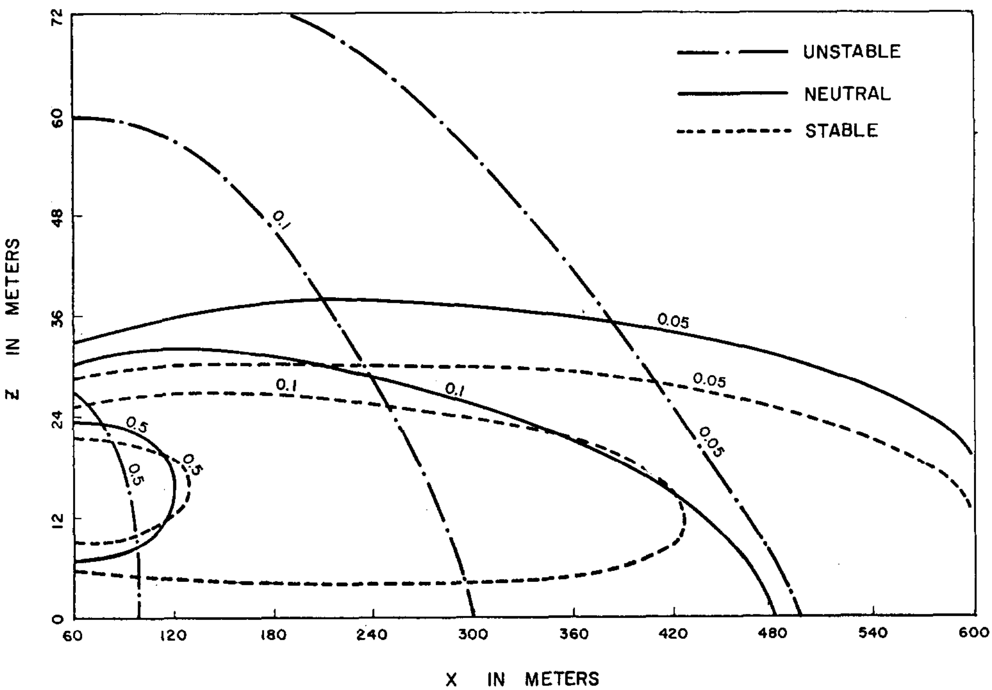

Fig. 2. Same as Figure 1 except that the point source is at $z_s=15$ m.

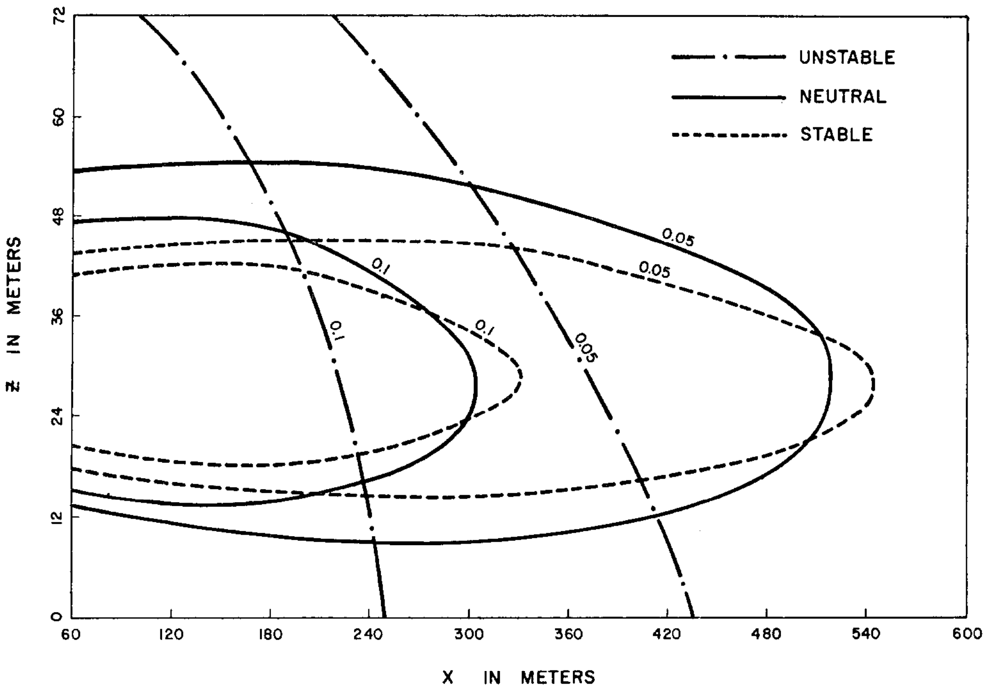

Fig. 3. Same as Figure 1 except that the point source is at $z_s = 30$ m.

TABLE I

Parameters used for three different atmospheric stability conditions

	z_0 cm (1)	u^* cm s^{-1} (2)	L m (3)	α (4)	C (5)	β (6)	a cm$^{1-\alpha}$ (7)	b cm$^{2-\beta}$ (8)	k m$^{2-\alpha-p}$ (9)	p (10)
Unstable	1.0	10.0	−17	0.049	7.4	0.951	74	27.6	1.72	0.667
Neutral	1.0	10.0	∞	0.145	6.1	0.855	61	11.3	1.65	0.667
Stable	1.0	10.0	333	0.246	3.3	0.745	33	12.3	1.58	0.667

Note:

(1) Columns (1), (2) and (3) are the hypothetical surface roughness, friction velocity and atmospheric stability conditions.

(2) Columns (4) and (5) are the values obtained from the tables of Brutsaert and Yeh (1970a).

(3) Columns (9) and (10) are the presumed hypothetical values for illustrative purposes only. These values of k and p require further research.

(4) Columns (6), (7) and (8) are the values calculated from Equation (20).

layer of the atmosphere result in the observed elongated plume in the longitudinal direction. The increase of the source height results in increasing the ratio of the plume width to the plume length although wind speed increases with height. The various sizes of eddies are in response to the plume spreading under various atmospheric conditions. Thus, the width to length ratio of the plume increases as the atmosphere becomes more unstable.

5. Conclusion

Analytical solution for a steady state, three-dimensional turbulent diffusion equation with variable coefficients, is obtained. The variations of wind speed and vertical diffusivity with height as specified by power laws are considered in the diffusion equation. In particular, the lateral diffusivity as a function of height and downstream distance is introduced. The Gaussian diffusion model and Yih's model (1952b) are special cases of the present model. Thus, the present model is a generalization of the three-dimensional turbulent diffusion equation.

The practical application for predicting gas concentrations is described. The methods of selecting the parameters are outlined. Illustrative examples indicate that the differences in the concentration patterns between the unstable condition and the neutral and stable conditions are very pronounced. An elongated plume is observed for a source close to the ground. Plume length for an elevated source is shorter than that for a source at a lower level, although wind speed increases with height. These consequences are in response to the size of the eddies in the vertical direction. The sizes of eddies vary with atmospheric stability, and hence the width to length ratio of a plume also depends on stability. The values of this ratio increases with decreasing stability and increases with increasing source height.

The combined exponential Bessel distribution of Equation (20) is superior logically to the currently adopted Gaussian diffusion model in air pollution modeling. The former is based on realistic power law approximations of the wind velocity and eddy diffusivities while the latter is based on the questionable assumptions of constant velocity and diffusivities. The non-Gaussian behavior of the concentration distribution under unstable atmospheric conditions can be obtained by the present model but not by the Gaussian diffusion model. The availability of this paper should obviate the need of using an unrealistic Gaussian distribution in air pollution modeling, especially in unstable atmospheric conditions.

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